



UNIVERSITY OF SCIENCE AND TECHNOLOGY IN ZEWAIL CITY

NANENG 205

PHYSICS OF SEMICONDUCTORS

Topic 2

Temperature (T) vs. Carrier Concentrations and V_{bi}

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Introduction

Our project introduces a study of Temperature (T) vs. Carrier Concentrations (n_n, n_p, p_p, p_n, n_i) and V_{bi} . It simulates an abrupt Silicon p-n junction and sweeps the temperature to observe its effects on carrier concentrations and built-in potential.

The values used in this project were:

- **Temperature**

- Individual Run: 400 K
- Sweep: 25 $\rightarrow$ 900 K

- **Donor Concentration**

- Individual Run: $1 \times 10^{18} \text{ /cm}^3$
- Sweep: $1 \times 10^{18} \text{ /cm}^3$

- **Acceptor Concentration**

- Individual Run: $1 \times 10^{17} \text{ /cm}^3$
- Sweep: $1 \times 10^{17} \text{ /cm}^3$

Simulation results

```
ATLAS> EXTRACT> init infile="Topic2LogFile.log"
EXTRACT> extract name="Majority Electron Concentration (/cm^3)" max(curve(v."anode", probe."nn"))
Majority Electron Concentration (/cm^3)=8.72508e+017
EXTRACT> extract name="Majority Hole Concentration (/cm^3)" max(curve(v."anode", probe."pp"))
Majority Hole Concentration (/cm^3)=9.21562e+016
EXTRACT> extract name="Minority Electron Concentration (/cm^3)" max(curve(v."anode", probe."np"))
Minority Electron Concentration (/cm^3)=4.53383e+008
EXTRACT> extract name="Minority Hole Concentration (/cm^3)" max(curve(v."anode", probe."pn"))
Minority Hole Concentration (/cm^3)=1.00291e+008
EXTRACT> extract name="Intrinsic Carrier Concentration (/cm^3)" max(curve(v."anode", sqrt(probe."nn"*probe."pn")))
Intrinsic Carrier Concentration (/cm^3)=9.35438e+012
EXTRACT> extract name="V_bi" max(curve(v."anode", probe."Max Potential"-probe."Min Potential"))
V_bi=0.724301
```

Figure 1: Results of an individual run

After that we did the sweep over a different temperature range and all results and data are provided in the zip file.

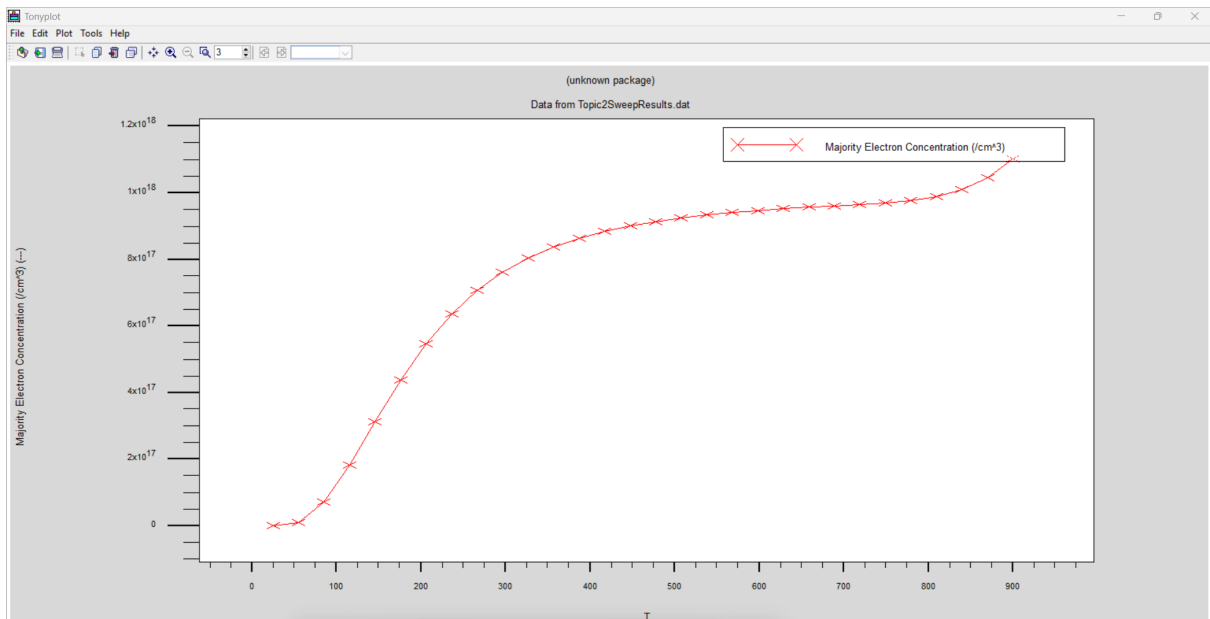


Figure 2: Majority electron concentration vs T

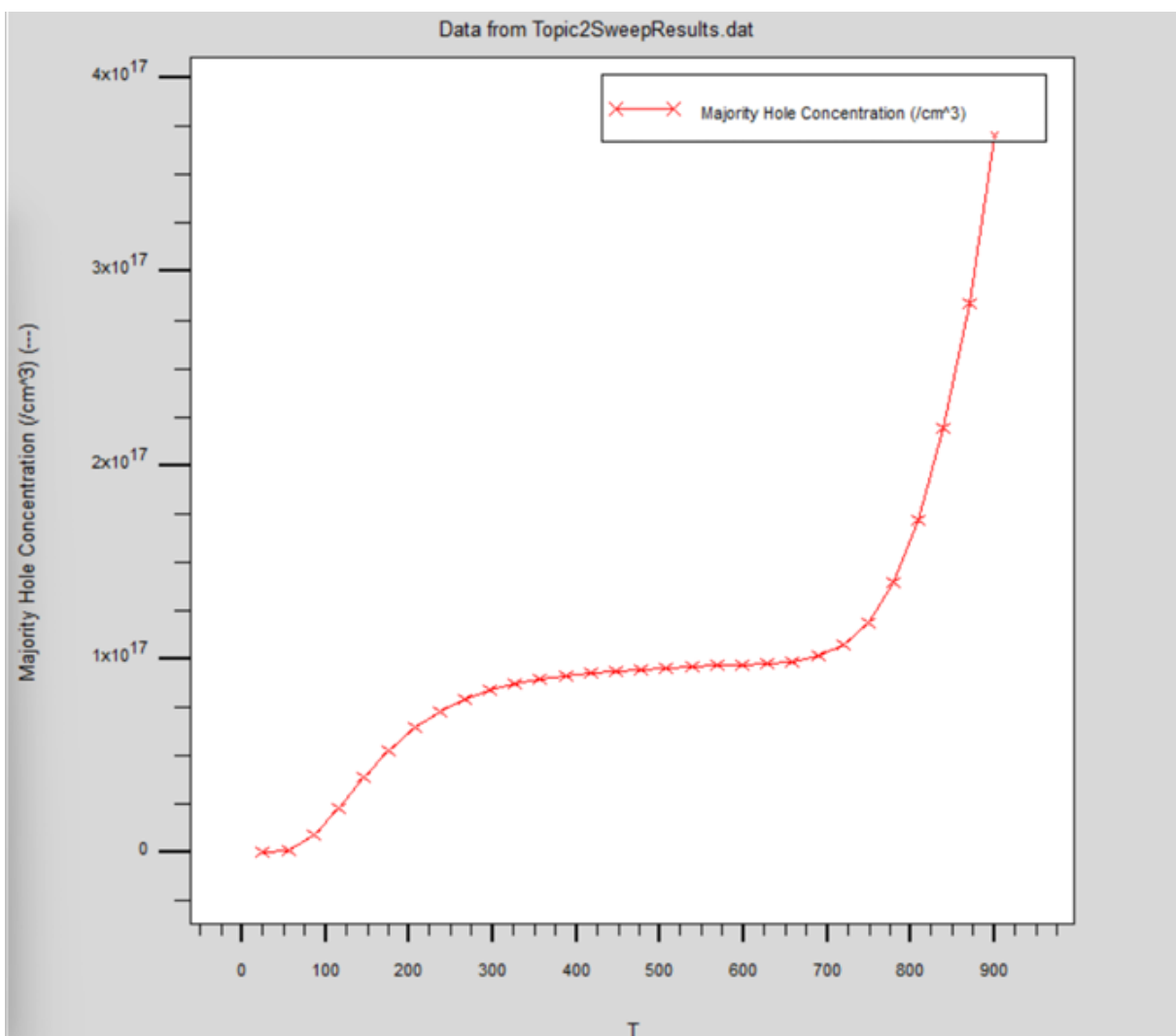


Figure 3: Majority hole concentration vs T

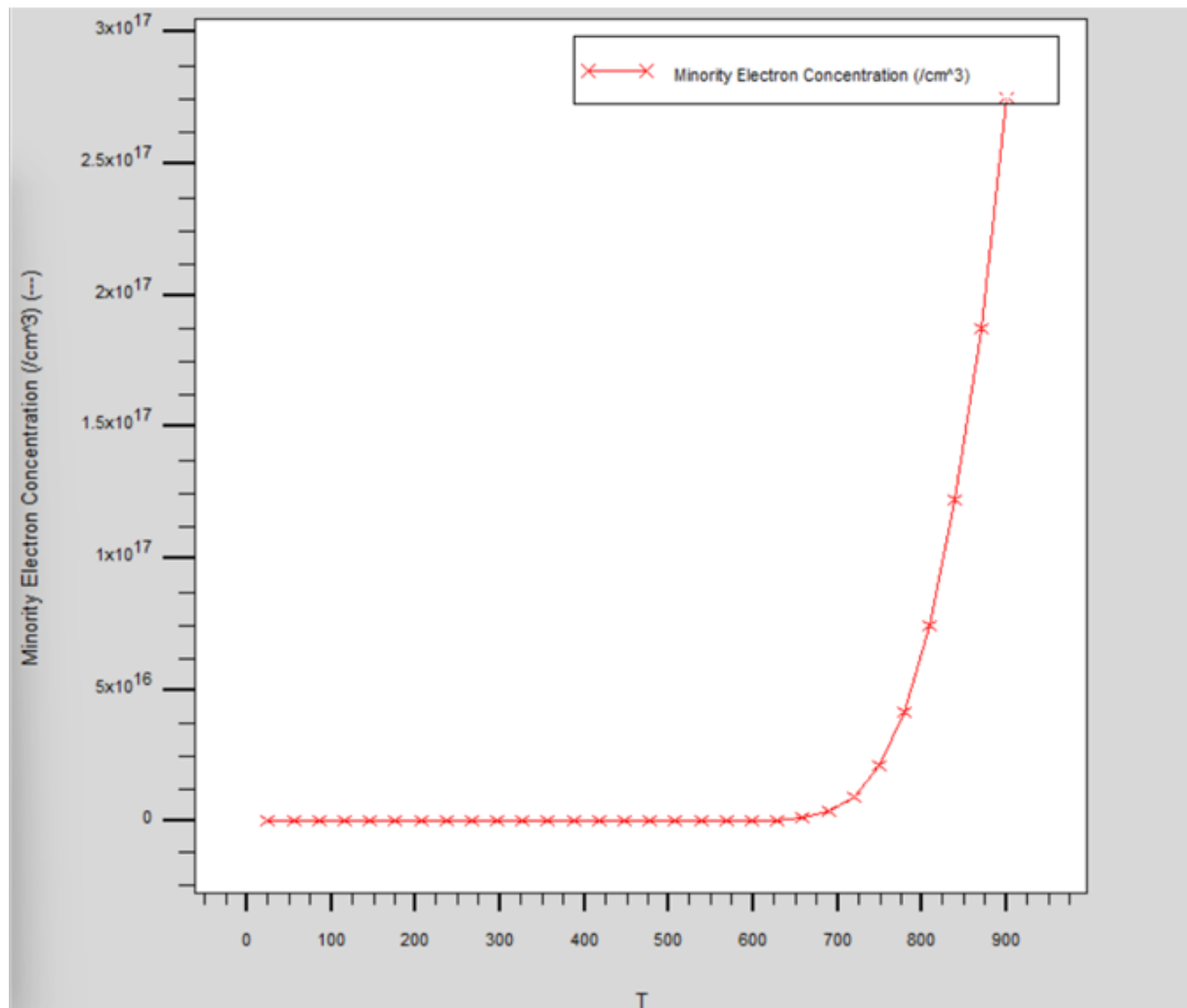


Figure 4: Minority electron concentration vs T

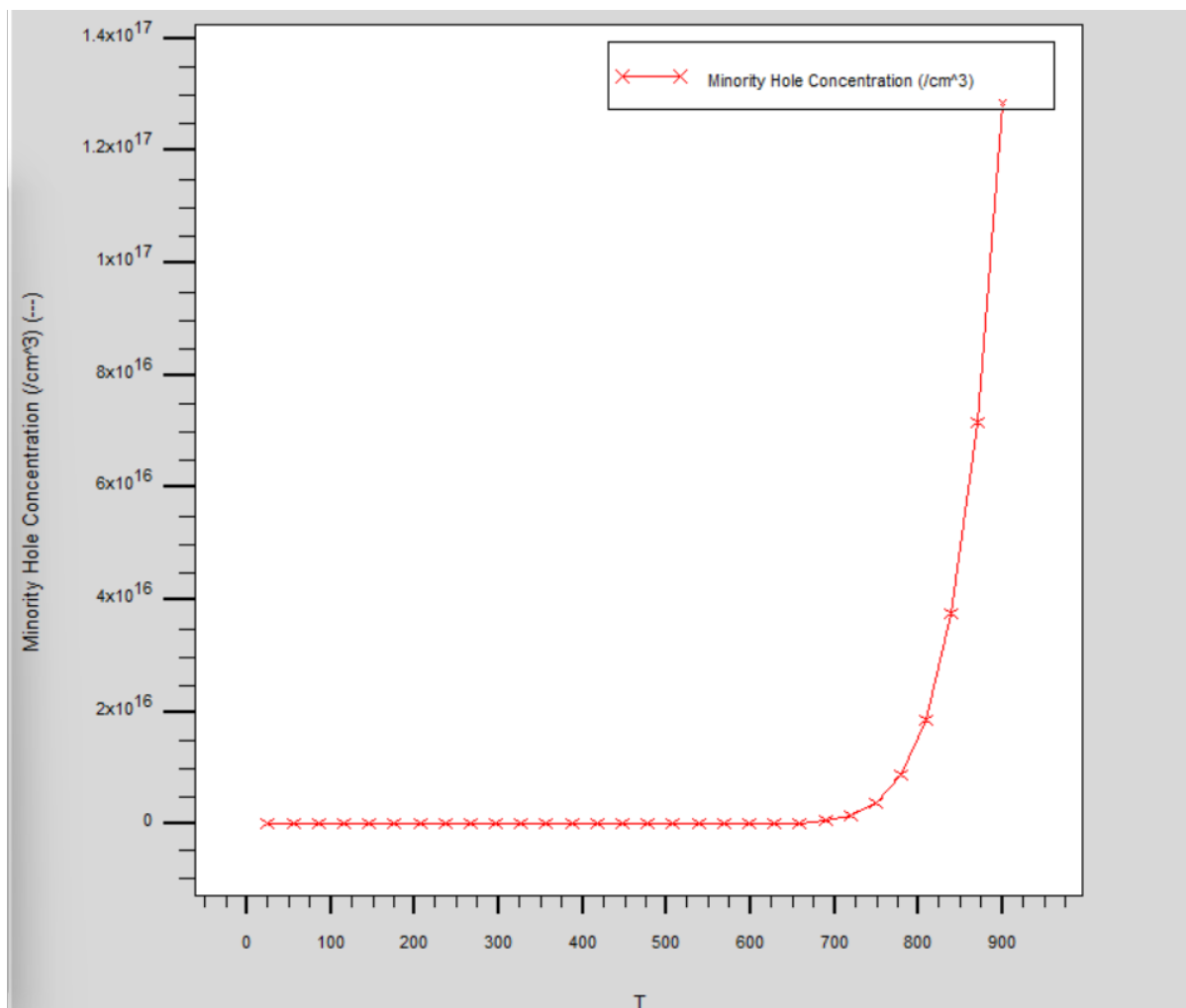


Figure 5: Minority hole concentration vs T

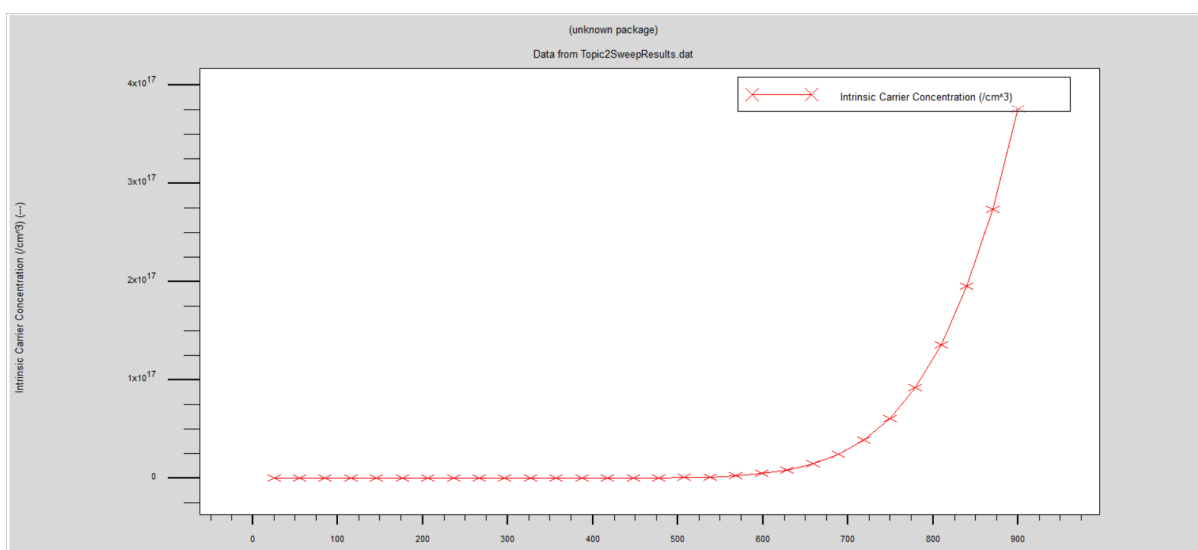
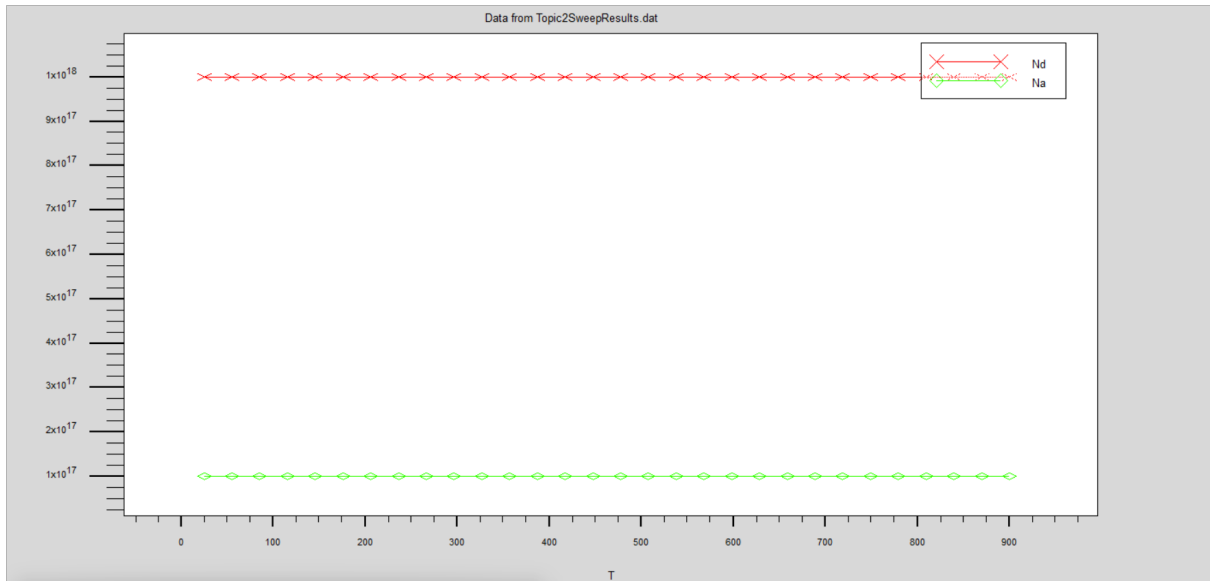
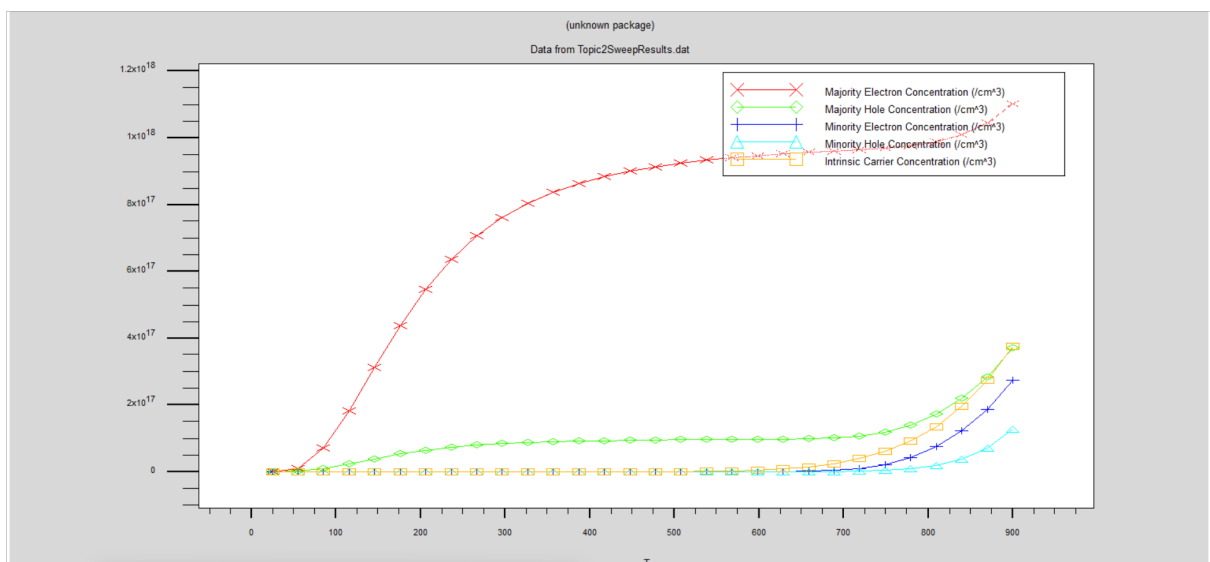
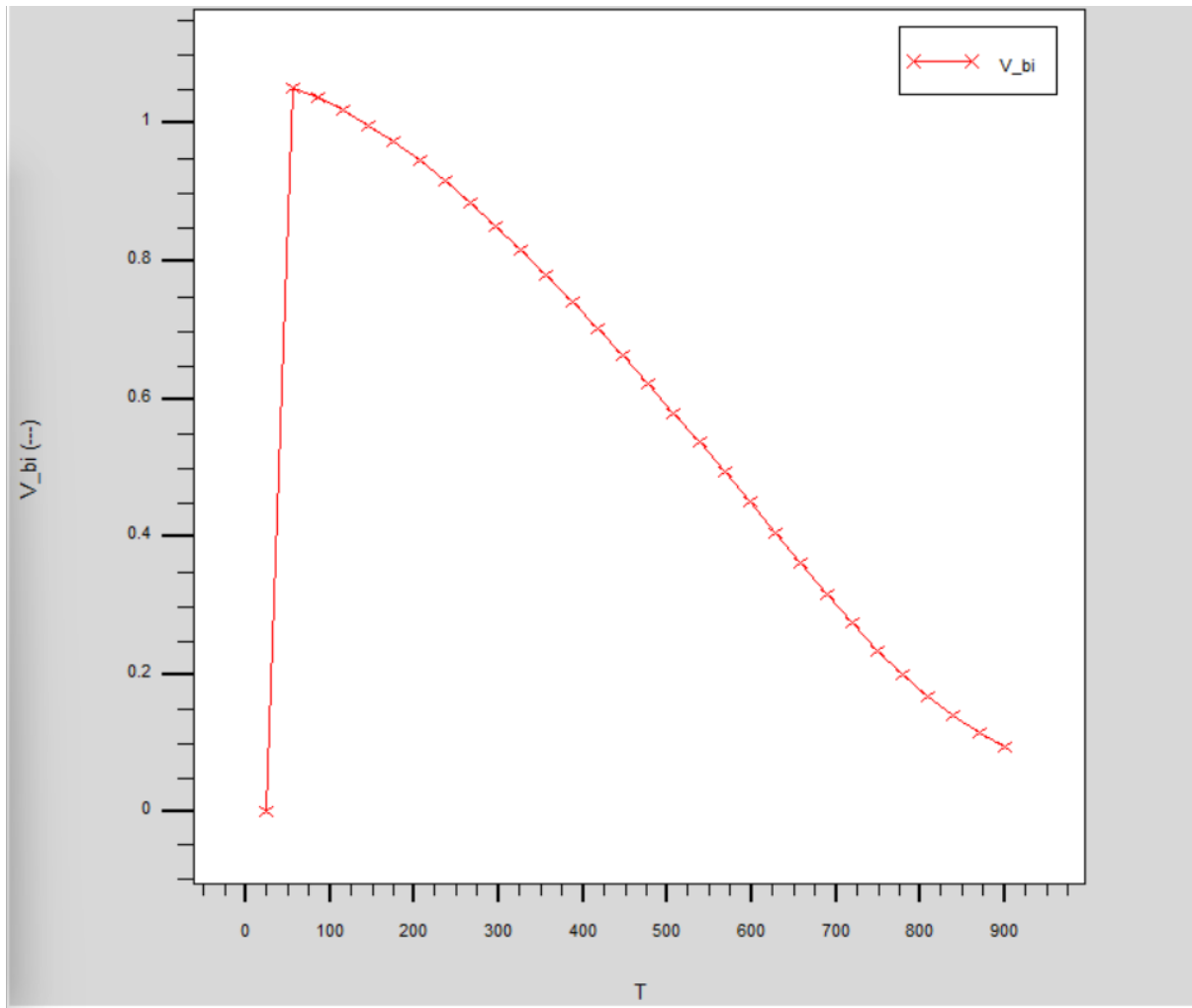


Figure 6: Intrinsic carrier concentration vs T

Figure 7: N_d and N_a vs T

Figure 8: V_{bi} vs T

Research Output and explanations

The similarity between these two figures below that majority electron concentration goes through four phases and what determines them is the temperature

- **First phase: freeze out**

- This happens over a low-temperature region.

- **Second phase: partial ionization**

- As the temperature increases, additional donor and acceptor atoms become thermally ionized, causing the free carrier concentration to increase. For partial ionization, the electron concentration is given by

$$n_0 = \frac{N_D}{1 + \frac{1}{2} \exp\left(\frac{E_D - E_F}{kT}\right)}$$

- **Third phase: Extrinsic**

- At room temperature at $T = 300\text{K}$ complete ionization of donor or acceptor atoms occurs. the majority carrier concentration is primarily determined by the dopant atoms, such that

$$n_0 \approx N_D$$

- **Forth phase: Intrinsic**

- In the intrinsic region, the temperature becomes sufficiently high that thermally generated electron-hole pairs dominate the behavior of the semiconductor. As the temperature increases, the intrinsic carrier concentration n_i rises rapidly and eventually exceeds the dopant concentration. In this stage, the carrier concentrations are mainly controlled by thermal generation rather than impurity atoms, and

$$n_0 \approx p_0 \approx n_i$$

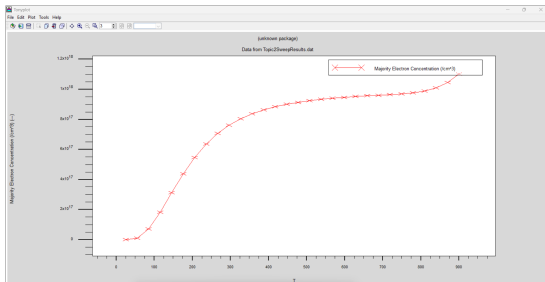


Figure 9: n_0 vs T

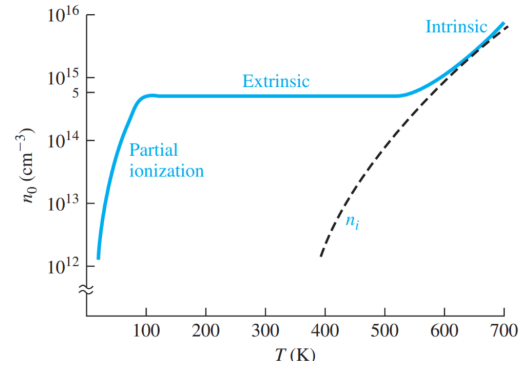


Figure 10: Reference graph

The behavior of V_{bi} with temperature is described by

$$V_{bi} = \frac{kT}{q} \ln \left(\frac{N_a N_d}{n_i^2} \right) \quad (1)$$

That's why at $T = 0\text{K}$ the value of V_{bi} is also zero, although one can think it increases with temperature linearly judging from their relation but this is not true because the exponential increase of n_i is far more significant than the linear increase of T , the overall built-in potential V_{bi} decreases as temperature rises.

The physical interpretation is that as temperature increases, the generation of intrinsic carriers causes the p-type and n-type regions to lose their distinct extrinsic characteristics and behave more like intrinsic silicon.

	Germanium	Silicon	Gallium Arsenide
$T = 300 \text{ K}$	0.66 eV	1.12 eV	1.42 eV
$T = 400 \text{ K}$	0.62 eV	1.09 eV	1.38 eV
$T = 500 \text{ K}$	0.58 eV	1.06 eV	1.33 eV
$T = 600 \text{ K}$	0.54 eV	1.03 eV	1.28 eV

Quantity	Low T	Medium T	High T
(n_i)	very small	moderate	huge
(n_n)	freeze-out	(= N_D)	intrinsic rise
(p_p)	freeze-out	(= N_A)	intrinsic rise
(n_p)	tiny	small	increases rapidly
(p_n)	tiny	small	increases rapidly
(V_{bi})	larger	normal	decreases

1-Intrinsic Carrier Concentration(n_i):

$$n_i = \sqrt{N_c N_v} e^{-\frac{E_g}{2kT}}$$

2-n-n Majority Electron Concentration in n-type

$$n_n = \frac{(N_D - N_A)}{2} + \sqrt{\left(\frac{N_D - N_A}{2}\right)^2 + n_i^2}$$

Approximation:

$n_n \approx N_D$ at room temperature

3-pn Minority Hole Concentration in n-type:

$$p_n = \frac{n_i^2}{n_n}$$

4-p-p Majority Hole Concentration in p-type:

$$p_p = \frac{(N_A - N_D)}{2} + \sqrt{\left(\frac{N_A - N_D}{2}\right)^2 + n_i^2}$$

Approximation:

$$p_p \approx N_A$$

5-n-p Minority Electron Concentration in p-type:

$$n_p = \frac{n_i^2}{p_p}$$

6-V_{bi} Built-in Voltage:

$$V_{bi} = \frac{kT}{q} \ln \left(\frac{N_A N_D}{n_i^2} \right)$$

7-Energy band gap:

$$E_g(T) = E_g(0) - \frac{\alpha T^2}{T + \beta}$$

For silicon :

- $E_g(0) = 1.17 \text{ eV}$
- $\alpha = 4.73 \times 10^{-4} \text{ eV/K}$
- $\beta = 636 \text{ K}$

Table 4.1 Effective density of states function and density of states effective mass values				
	$N_c \text{ (cm}^{-3}\text{)}$	$N_v \text{ (cm}^{-3}\text{)}$	m_n^*/m_0	m_p^*/m_0
Silicon	2.8×10^{19}	1.04×10^{19}	1.08	0.56
Gallium arsenide	4.7×10^{17}	7.0×10^{18}	0.067	0.48
Germanium	1.04×10^{19}	6.0×10^{18}	0.55	0.37

From table:

$$N_c(300) = 2.8 \times 10^{19} \text{ cm}^{-3}$$

$$N_v(300) = 1.04 \times 10^{19} \text{ cm}^{-3}$$

at any T

$$N_c(T) = N_c(300) \left(\frac{T}{300} \right)^{3/2}$$

$$N_v(T) = N_v(300) \left(\frac{T}{300} \right)^{3/2}$$

At T=900K:

the bandgap at 900K is approximately:

$$E_g \approx 0.92 \text{ eV}$$

$$N_c(900) \approx 1.46 \times 10^{20} \text{ cm}^{-3}$$

$$N_v(900) \approx 5.40 \times 10^{19} \text{ cm}^{-3}$$

So

$$n_i = \sqrt{(1.46 \times 10^{20})(5.40 \times 10^{19})} e^{-\frac{0.92}{2(8.617 \times 10^{-5})(900)}}$$

$$n_i \approx 2.36 \times 10^{17} \text{ cm}^{-3}$$

$$N_A = 10^{17} \text{ cm}^{-3}$$

$$N_D - N_A = 9 \times 10^{17} \text{ cm}^{-3}$$

1) Majority Electron Concentration n_n

$$= 4.5 \times 10^{17} + \sqrt{(4.5 \times 10^{17})^2 + (2.34 \times 10^{17})^2}$$

$$N_n = 9.57 \times 10^{17} \text{ cm}^{-3}$$

2) Minority Hole Concentration p_n

$$= \frac{(2.34 \times 10^{17})^2}{9.57 \times 10^{17}}$$

$$p_n = 5.72 \times 10^{16} \text{ cm}^{-3}$$

3) Majority Hole Concentration p_p

$$p_p = \frac{N_A - N_D}{2} + \sqrt{\left(\frac{N_A - N_D}{2}\right)^2 + n_i^2}$$

$$= -4.5 \times 10^{17} + 5.07 \times 10^{17}$$

$$p_p = 5.7 \times 10^{16} \text{ cm}^{-3}$$

4) Minority Electron Concentration n_p

$$n_p = 9.61 \times 10^{17} \text{ cm}^{-3}$$

5) Built-in Voltage V_{bi}

$$V_{bi} = 0.0777 \ln \left(\frac{(10^{17})(10^{18})}{(2.34 \times 10^{17})^2} \right)$$

$$V_{bi} = 0.047 \text{ V}$$

Majority Electron Concentration (/cm³)=1.10237e+018

Majority Hole Concentration (/cm³)=3.70221e+017

Minority Electron Concentration (/cm³)=2.74899e+017

Minority Hole Concentration (/cm³)=1.28467e+017

Intrinsic Carrier Concentration (/cm³)=3.76322e+017

V_{bi}=0.0950386

The analytical textbook calculation for the intrinsic carrier concentration at 900 K approximately $2.36 \times 10^{17} \text{ cm}^{-3}$, whereas the Silvaco ATLAS simulation predicts a higher value of $3.76 \times 10^{17} \text{ cm}^{-3}$. This discrepancy is physically justified by the advanced physical models enabled in the simulation environment.

The simplified analytical model assumes constant effective carrier masses and accounts for bandgap variation only through the standard temperature-dependent Varshni relation. In contrast, Silvaco ATLAS incorporates more sophisticated semiconductor

physics, including temperature-dependent effective masses and Klaassen Bandgap Narrowing (BGN.KLAASSEN).

Due to the relatively high donor concentration (10^{18} cm^{-3}), additional bandgap narrowing occurs inside the silicon material. This effectively reduces the energy required for thermal carrier generation, thereby increasing the intrinsic carrier concentration. Consequently, the Silvaco result is expected to be larger and more physically accurate than the simplified hand calculation

Silvaco constants :

CONSTANTS:

```
Boltzmann's constant = 1.38066e-023 J/K
Elementary charge    = 1.60219e-019 C
Permittivity in vacuum = 8.85419e-014 F/cm
Temperature           = 400 K
Thermal voltage       = 0.0344694 V
```

REGIONAL MATERIAL PARAMETERS:

```
Region      :      1      2      3
Material    :      Silicon Conductor Conductor
Type        :      semicond. metal metal
```

At T=400K

Band Parameters

```
Epsilon      :      11.8
Eg (eV)      :      1.05
Chi (eV)     :      4.18
Nc (per cc)  :      4.31e+019
Nv (per cc)  :      1.6e+019
ni (per cc)  :      6.16e+012
```

At T=400K calculations:

$$E_g(400) = 1.17 - \frac{(4.73 \times 10^{-4})(400)^2}{400 + 636}$$

$E_g(400) \approx 1.097 \text{ eV}$

Silvaco $E_g = \text{Baseline} - \text{Thermal Shift} - \text{Doping Shift (BGN)}$

$$E_g = 1.17 \text{ eV} - 0.073 \text{ eV} - 0.047 \text{ eV} = \mathbf{1.05 \text{ eV}}$$

That change in E_g which made it more narrow

2) Effective Density of States

$$\begin{aligned} N_c(400) &= 2.8 \times 10^{19} \left(\frac{400}{300} \right)^{3/2} \\ &= 2.8 \times 10^{19} (1.539) \\ N_c(400) &\approx 4.31 \times 10^{19} \text{ cm}^{-3} \end{aligned}$$

$$N_v(400) \approx 1.60 \times 10^{19} \text{ cm}^{-3}$$

N_c and N_v is like silvaco , so change only in E_g